

First Order Equation

$$\tau \frac{dT}{dt} + T = T_{\infty}$$

$$\frac{dT}{dt} + \frac{1}{\tau} T = \frac{1}{\tau} T_{\infty}$$

Solution

$$T(t) = T_0 e^{-\frac{t}{\tau}} + T_{\infty} \left[1 - e^{-\frac{t}{\tau}} \right]$$

Transfer Function

$$F(s) = \frac{T_{\infty}}{\tau} \cdot \frac{1}{s \left(s + \frac{1}{\tau} \right)}$$